Earnings Call Sentiment and Realized Volatility Prediction

A Fine-Tuned FinBERT Approach on S&P 500, 2016–2025

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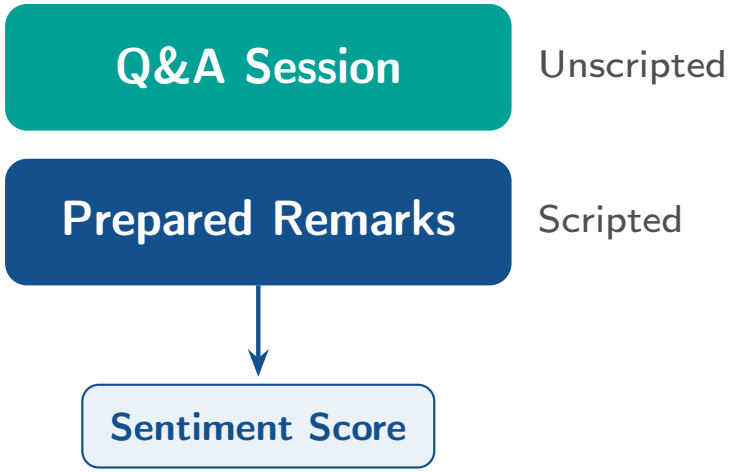
Why Earnings Calls Carry Financial Signal

- Each call ≈ **8,000 words**, two segments:
  - **Q&A** unscripted, reactive
  - **Prepared remarks** scripted, forward-looking
- Hard numbers alone don't tell the full story
- **Tone** conveys soft information: confidence, uncertainty, strategic framing
- Investors react to *how* things are said, not just *what*



Key References

Price et al. (2012): tone predicts abnormal returns beyond earnings surprise
Hassan et al. (2019): political-risk language in calls predicts stock volatility

Research Question

*Does the **cautious tone** of management's prepared remarks in S&P 500 earnings calls predict **realized stock volatility** over the 20 trading days following the call, beyond historical volatility and the VIX?*

Step 1	Step 2	Step 3
Knowledge Distillation (financial_phrasebank + Groq/Llama)	FinBERT Fine-Tuning	OLS Regression with Fixed Effects

Data: From Raw Transcripts to Analysis-Ready Panel

Transcript Corpus (Koyfin)

Source	Koyfin (Selenium)
Raw corpus	113,990 transcripts
After S&P filter	**14,963 transcripts**
Firms covered	**488 S&P 500 firms**
Period	2016–2025
Sectors	11 GICS sectors

Finance Panel (Yahoo Finance)

Observations	**14,847**
Tickers	486
`rv_post` mean	0.274
`rv_hist` mean	0.293
VIX mean	18.99
Attrition rate	$< 1\%$

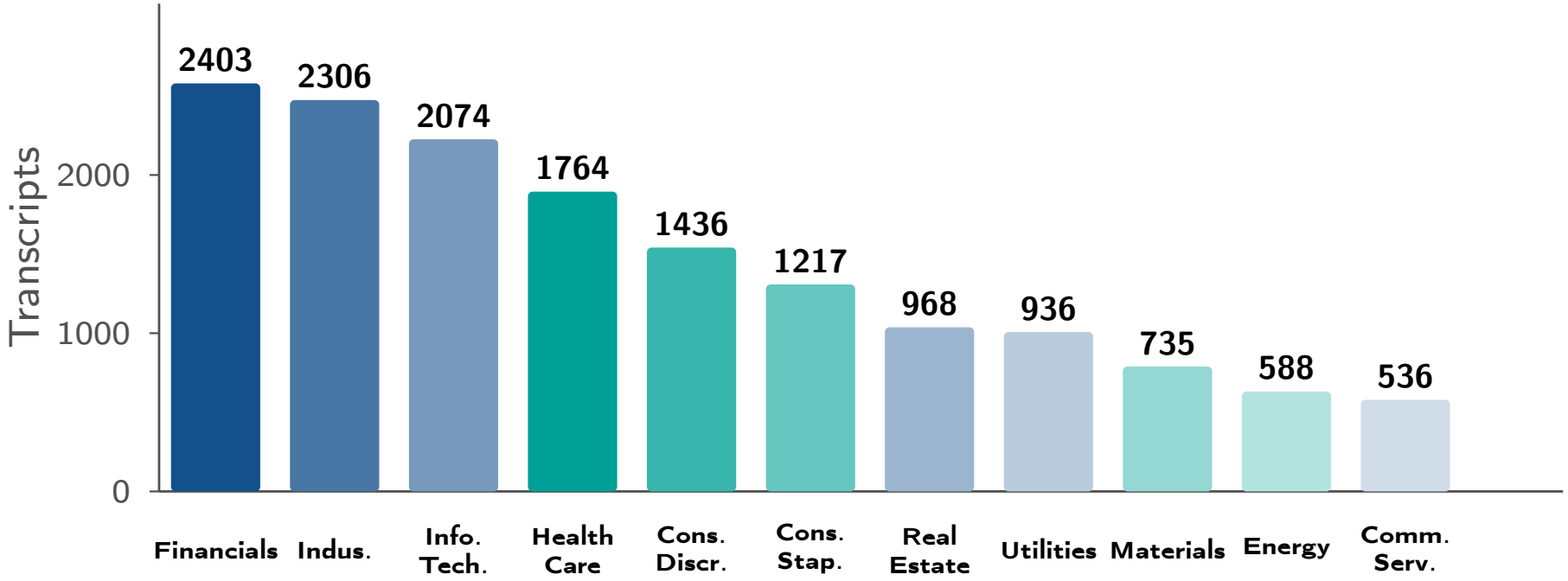
Entity Matching Challenge

"Google" = "Alphabet" = "Alphabet Inc."
Canonical alias table required before any NLP step

Realized Volatility

$\texttt{rv_post} = \hat{\sigma}_{\text{daily}} \times \sqrt{252}$
Days $t{+}1$ to $t{+}21$ **no look-ahead bias**

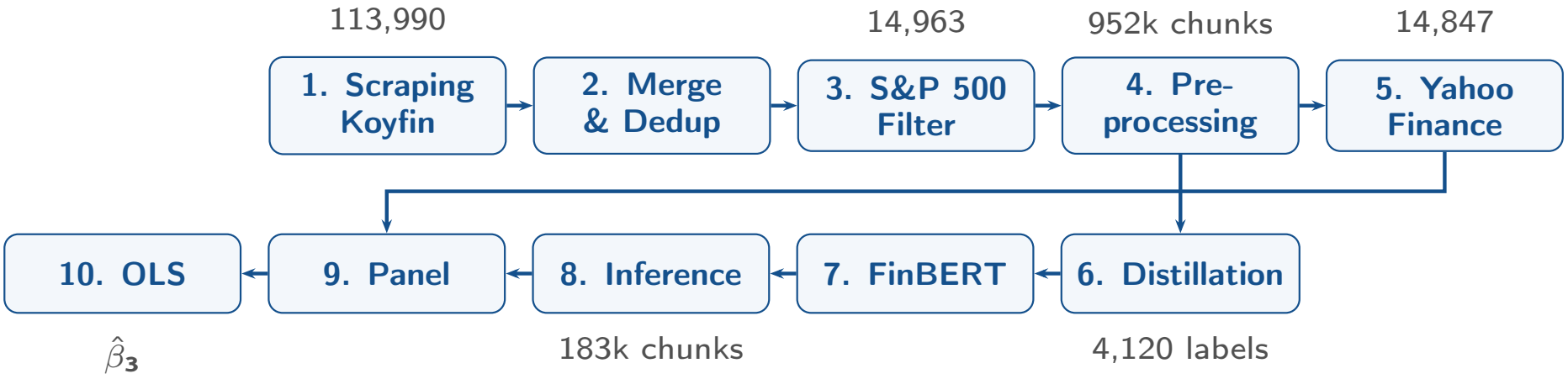
Sample Coverage Across 11 GICS Sectors



Interpretation

Broad cross-sectional coverage across all 11 GICS sectors. Financials, Industrials, and Information Technology together account for more than 45% of the final sample, limiting the risk of industry-specific confounds.

End-to-End Pipeline



Text Preprocessing: 6-Stage Pipeline

1. **Metadata normalization** ticker, sector, date, quarter
2. **Speaker segmentation** via regex
   → management / analyst / operator
3. **Section labeling** via cumsum:
   `prepared_remarks` vs `Q&A`
4. **Text cleaning** HTML, operator noise
5. **Sentence-level chunking**
   never split across speaker turns
6. **Token verification** with FinBERT tokenizer
   (max 510 tokens *not* whitespace count)

Design rule

No stemming, no stopword removal.
Negations carry sentiment signal for BERT.

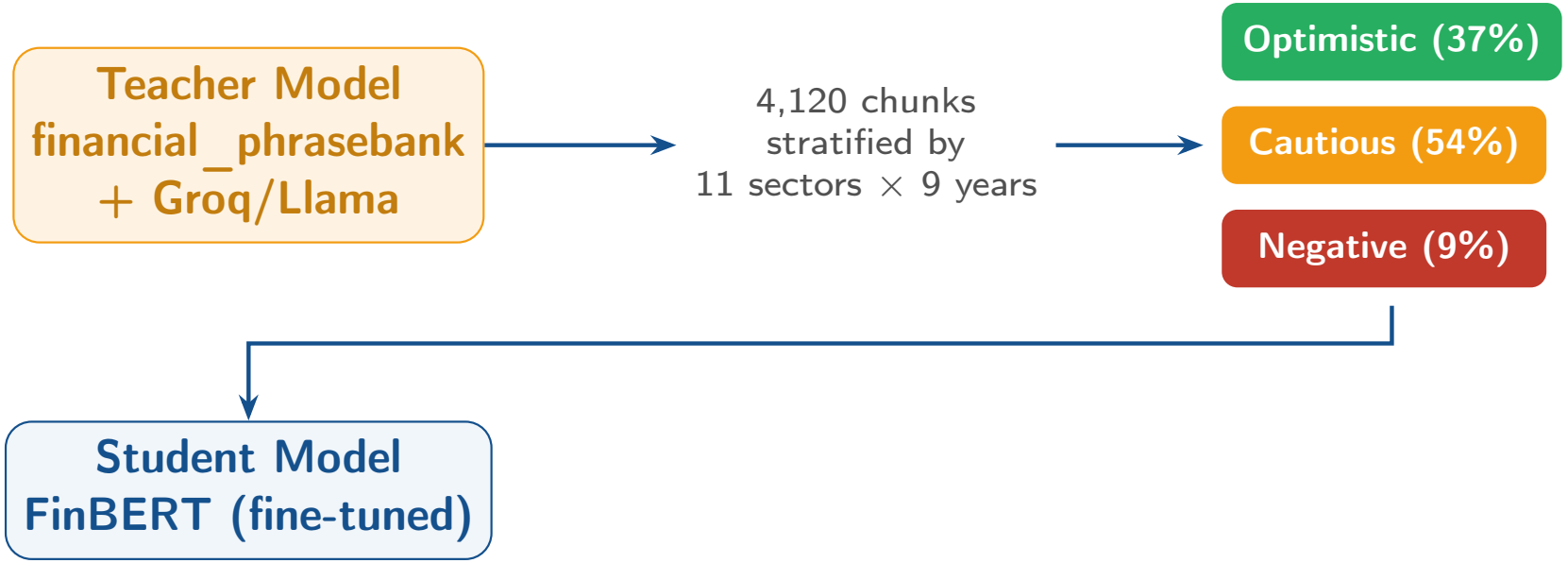
Preprocessing Output

Metric	Value
Total chunks	952,520
Avg tokens/chunk	174.9
Max tokens/chunk	510
Main corpus (mgmt + prep.)	
Chunks retained	183,491
Share of total	19.3%

Why prepared remarks only?

Scripted, forward-looking, directly controlled by management more informative for predicting volatility.

Knowledge Distillation: Building the Training Set



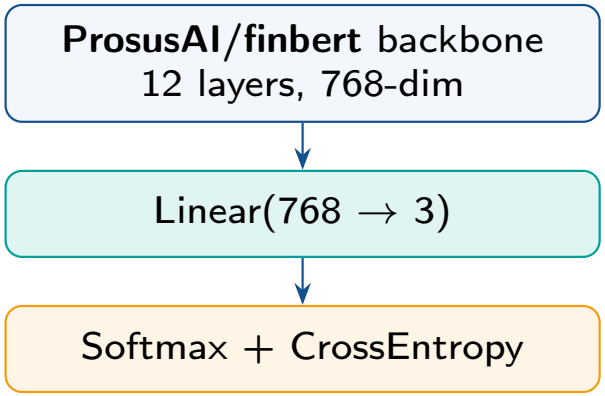
LLM labels: `temperature=0`, deterministic

Why stratified sampling?

Ensures label coverage across all 11 GICS sectors and all years 2016–2025, avoiding concentration in any single sector or time period.

FinBERT Fine-Tuning

Model architecture:



Why Softmax + CrossEntropy?
Each chunk belongs to exactly *one* class
mutually exclusive by construction.

Training Setup

Optimizer	AdamW
LR	2×10^{-5}
Epochs	3
Batch size	16
Split	80 / 10 / 10
Stratified	sector × year
Hardware	Google Colab T4 GPU

Output Variable

$$\mathrm{cs}_{i,t} = \frac{1}{|K|} \sum_{k \in K} \hat{P}(\text{cautious} \mid x_k)$$

One score $\in [0, 1]$ per call.
Mean: 0.399 Std: 0.141

Econometric Design

Two-model comparison:

Baseline model

$$\texttt{rv_post}_{i,t} = \alpha + \beta_1 \, \texttt{rv_hist}_{i,t} + \beta_2 \, \texttt{vix}_t + \delta_s + \tau_q + \varepsilon_{i,t}$$

Full model

$$\texttt{rv_post}_{i,t} = \alpha + \beta_1 \, \texttt{rv_hist}_{i,t} + \beta_2 \, \texttt{vix}_t + \beta_3 \, \texttt{cautious_share}_{i,t} + \delta_s + \tau_q + \varepsilon_{i,t}$$

Fixed effects:

- δ_s sector FE: absorbs cross-sectional vol heterogeneity
- τ_q quarter FE: captures earnings seasonality

Identification safeguards:

- `rv_post` starts at $t+1$ **no look-ahead bias**
- HC3 robust standard errors
- VIF: all regressors < 2

Test: Is $\hat{\beta}_3$ significant? Does $R^2_{\text{full}} > R^2_{\text{baseline}}$?

FinBERT Classification Performance (Test Set)

Classification Report (412 obs.)

Class	Prec.	Rec.	F1	N
Optimistic	0.87	0.76	0.81	152
Cautious	0.85	0.91	0.88	221
Negative	0.86	0.92	0.89	39
Macro avg	0.86	0.87	0.86	412

Accuracy: 86% (3 epochs, T4 GPU)

Key observations

Cautious class: F1 = 0.88
Recall = 0.91 the model detects cautious chunks with high coverage

This matters: `cautious_share` is our key regression variable

Training data

Cautious: 53.6% (2,210)
Optimistic: 36.9% (1,522)
Negative: 9.4% (388)
LLM labels: temperature=0

Regression Results

	(1) Baseline	(2) Full Model
`rv_hist`	0.4266***	0.4265***
	(0.019)	(0.019)
`vix`	0.0040***	0.0040***
	(0.0005)	(0.0005)
`covid`	0.0815***	0.0815***
	(0.013)	(0.013)
`rate_hike`	0.0259***	0.0259***
	(0.006)	(0.006)
`cautious_share` ($\hat{\beta}_3$)		0.0007
		(0.010)
Sector FE	Yes	Yes
Quarter FE	Yes	Yes
Crisis FE	Yes	Yes
R^2	0.445	0.445
ΔR^2		0.000
N	10,226	10,226

HC3-robust standard errors in parentheses. $^{***}p < 0.001$.

Main Finding

A 10 percentage-point increase in ***cautious_share***

is associated with a ***0.007%*** *change in realized volatility over the 20 trading days following the call.*

$$\hat{\beta}_3 = 0.0007 \quad p = 0.945 \quad 95\%\ CI\ [-0.019,\ 0.021]$$

Not statistically significant.

What drives R^2 from 0.437 to 0.445?

Crisis dummies, not tone:
COVID-19: +8.2 pp volatility
Rate hike cycle: +2.6 pp volatility

Interpretation

Cautious language in scripted prepared remarks appears to be **efficiently priced** by markets — no residual signal in post-call vol.

Robustness Checks

Variance Inflation Factors

Variable	VIF
`cautious_share`	1.045
`rv_hist`	1.331
`vix`	1.288

All VIF $< 2 \Rightarrow$ **no multicollinearity**

Incremental F-test

H_0: $\beta_3 = 0$ (tone adds no power)
$F = 0.468$, $p = 0.494$
Fail to reject H_0

Tone Score Distribution

	cautious	optimistic
Mean	0.399	0.583
Std	0.141	0.145
P25	0.294	0.492
Median	0.383	0.600
P75	0.488	0.691

15,039 calls scored
478 S&P 500 firms

Discussion & Conclusion

Contributions:

1. 113,990-transcript pipeline with canonical entity matching
2. Domain-adapted FinBERT via knowledge distillation (4,120 labels)
3. Clean causal design: 20-day post-call window, HC3 SE, sector & quarter FE

Limitations:

- LLM labels may introduce noise on ambiguous chunks
- No control for earnings surprise (EPS vs. consensus)
- S&P 500 only $\Rightarrow$ survivorship bias

Main finding:

Cautious tone is **not** a significant predictor of post-call realized volatility ($\hat{\beta}_3 = 0.0007$, $p = 0.945$). Scripted management language appears to be efficiently priced in large-cap markets consistent with the view that prepared remarks are legally reviewed and strategically framed, leaving no residual signal.

Extensions:

- Q&A section: unscripted analyst tone
- Add earnings surprise control
- Non-US markets and smaller firms

NLP pipelines can extract risk-relevant soft information from earnings calls that structured financial databases do not capture.

Thank you

Questions welcome

Data	Koyfin transcripts, Yahoo Finance, CBOE VIX
Models	ProsusAI/finbert, Groq/Llama-3.1-8B
Code	Python, Selenium, HuggingFace datasets, Transformers, statsmodels